Naeim Mousavi

Applied Scientist | Machine Learning Scientist | Data Scientist

Madrid, Spain

GitHub Profile: https://github.com/nmousavi2020/
LinkedIn Profile: https://www.linkedin.com/in/naeim-mousavi/



Skills

Machine Learning
Gradient Boosting
Deep Learning
Neural Networks

Data & Scientific Computing
Geospatial & Satellite Data
Signal Processing
Scientific Computing

Programming
Python
(NumPy, pandas, scikit-learn)
PyTorch

ML Engineering & MLOps
MLflow
FastAPI
Docker

Profile

Applied Machine Learning Scientist and Data Scientist with 6 years of experience designing scalable ML systems for complex, large-scale real-world problems. Specialized in multi-source data integration, feature engineering, uncertainty quantification, model development and deployment. Skilled in Python, PyTorch, gradient boosting, and MLOps tools including MLflow and Docker. Experienced in building scalable, interpretable, and deployable ML systems.

Selected Project

MLOps_Churn_Prediction | 2025-2026
Built and deployed a 7-stage end-to-end MLOps pipeline for customer churn prediction using Python, XGBoost, MLflow, FastAPI, and Docker. Compared two classification models on a synthetic dataset with 9 customer features, implemented MLflow experiment tracking, deployed the selected model as a REST API, and added automated API testing with 100% test pass rate.

DocIntel AI — LLM-Powered Enterprise Document Intelligence| 2025-2026
Developed a privacy-focused, LLM-powered enterprise document intelligence platform using hybrid RAG, BGE-M3 embeddings, BM25, Qdrant, BGE reranking, and locally hosted Qwen 2.5, enabling citation-grounded PDF question answering and structured information extraction while keeping sensitive documents and LLM inference within the local environment.

Professional Experience

Chinese Academy of Sciences & Complutense University of Madrid | China & Spain
Machine Learning Research Scientist | 2024 – 2026

- Developed PyTorch-based CNN models to infer subsurface properties beneath the oceans from satellite data, reducing computation time from 12 hours to 20 minutes (97%) for large-scale prediction.
- Designed a scalable machine learning framework for global prediction using feature engineering, hyperparameter optimization (from 22 to 6), and uncertainty quantification, producing interpretable models for large-scale applications.

Education

University of Kiel (CAU), Germany | 2016-2019
Doctorate in Natural Sciences (PhD), the Faculty of Mathematics and Natural Sciences

Selected Publication

1. Mousavi et al. 2026. Optimizing Feature Selection for Global Geothermal Heat Flow Prediction using Machine Learning. **Scientific Reports** https://doi.org/10.1038/s41598-026-63610-z
2. Mousavi et al. 2026. A Machine Learning Approach for Volcanic Eruption Mass Estimation. **Journal of Geophysical Research: Machine Learning and Computation** https://doi.org/10.1029/2026JH001264
3. Mousavi et al. 2026. Deep learning 3D inversion of gravity data at volcanic seamounts: An example of Conception Bank (Canary Islands). **Earth Science Informatics** https://doi.org/10.1007/s12145-026-02148-0